

Equity Buffer Preservation and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Buffer Preservation (EBP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EBP achieves an annualized gross (net) Sharpe ratio of 0.52 (0.43), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (18) bps/month with a t-statistic of 2.96 (2.44), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Inventory Growth, Investment to revenue, Accruals, Book leverage (annual)) is 17 bps/month with a t-statistic of 2.38.

1 Introduction

Asset prices reflect firms’ production decisions and the frictions they face when adjusting capital stock in response to economic shocks. The cross-section of expected returns varies systematically with characteristics that proxy for firms’ exposure to aggregate productivity risk and their ability to respond to investment opportunities [Cochrane and Piazzesi \(2005\)](#); [Zhang \(2005\)](#). We examine a novel predictor of stock returns based on firms’ preservation of equity buffers—the tendency to maintain financial flexibility through conservative capital structure choices. This behavior reflects fundamental economic trade-offs between current investment opportunities and the option value of preserving financial capacity for future productive investments [Bolton et al. \(2011\)](#). Despite extensive research on how financial constraints and capital structure decisions affect asset prices, the literature has not fully explored how firms’ active preservation of equity capacity relates to their risk exposures and expected returns in a production-based framework.

In a production economy, firms face convex adjustment costs when altering their capital stock, creating a wedge between the marginal product of capital and its user cost [Cooper and Haltiwanger \(2006\)](#). Equity Buffer Preservation (EBP) captures firms’ strategic decisions to maintain financial flexibility, which directly affects their ability to respond to productivity shocks. When firms preserve equity buffers by limiting leverage and maintaining conservative capital structures, they reduce their exposure to financial frictions that amplify adjustment costs during negative productivity realizations [Gomes and Schmid \(2010\)](#). This behavior is particularly valuable when investment opportunities are time-varying and firms face uncertainty about future productivity states.

The production-based asset pricing framework predicts that firms with higher EBP should command different risk premia because their investment policies and operating leverage differ systematically from firms that exhaust their equity capac-

ity [Zhang \(2005\)](#); [Liu et al. \(2009\)](#). Firms preserving equity buffers maintain lower operating leverage and face smaller costs of adjusting production in response to aggregate shocks. These firms can more efficiently reallocate resources when productivity conditions change, reducing their systematic risk exposure [Barro and Ursúa \(2008\)](#). The marginal value of installed capital for high-EBP firms is less sensitive to aggregate productivity fluctuations because these firms retain the flexibility to adjust their production scale without incurring prohibitive adjustment costs.

Furthermore, EBP proxies for firms' exposure to investment-specific technology shocks that affect the transformation of investment goods into productive capital [Papanikolaou and Kogan \(2014\)](#). Firms maintaining equity buffers can better exploit positive investment-specific shocks by rapidly scaling production when the relative price of investment goods falls. This optionality embedded in preserved equity capacity should be reflected in equilibrium asset prices through lower expected returns, as investors require smaller risk premia for holding claims on firms with greater operational flexibility [Carlson et al. \(2004\)](#). The cross-sectional variation in EBP thus maps directly to differences in firms' conditional risk premia through their differential exposures to both aggregate productivity risk and investment frictions.

We find strong evidence that Equity Buffer Preservation predicts the cross-section of stock returns. A value-weighted long/short trading strategy based on EBP achieves an annualized gross Sharpe ratio of 0.52, with the long-short portfolio earning average returns of 28 basis points per month (t-statistic = 3.77). The strategy's performance remains robust after controlling for established risk factors, generating a monthly alpha of 22 basis points (t-statistic = 2.96) relative to the Fama-French six-factor model. These results demonstrate significant economic magnitude, as a dollar invested in the EBP strategy would have grown to

3.57 over our sample period, ranking in the top 6

The predictive power of EBP persists across different portfolio construction methodologies and firm size categories. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the EBP strategy achieves average returns of 29 basis points per month (t -statistic = 3.28), with six-factor alphas of 21 basis points monthly. After accounting for transaction costs using high-frequency effective spreads, the strategy maintains a net Sharpe ratio of 0.43, outperforming 97

Controlling for the six most closely related anomalies from the factor zoo—including measures of financing activities, inventory growth, and leverage—the EBP strategy retains significant predictive power with a monthly alpha of 17 basis points (t -statistic = 2.38). The strategy loads significantly on the CMA (investment) factor with a coefficient of 0.23 (t -statistic = 4.52), consistent with EBP capturing variation in firms’ investment frictions and their ability to respond to productivity shocks. These results confirm that EBP contains unique information about cross-sectional return variation not subsumed by existing predictors, supporting its interpretation as a distinct proxy for production-based risk exposures.

Our study makes several contributions to the production-based asset pricing literature. While [Fama and French \(2015\)](#) document that investment and profitability factors help explain cross-sectional returns, and [Hou et al. \(2015\)](#) develop a q -factor model based on investment theory, our work identifies a novel characteristic that captures firms’ forward-looking preservation of financial flexibility within the production framework. Unlike static measures of leverage examined by [Penman et al. \(2007\)](#) or financing activities studied by [Bradshaw et al. \(2006\)](#), EBP reflects dynamic decisions about maintaining operational flexibility in the face of adjustment costs and productivity uncertainty.

Methodologically, we employ the comprehensive testing protocol of [Novy-Marx and Velikov \(2023\)](#) to establish the robustness of EBP’s predictive power, address-

ing concerns about data mining and implementation costs that plague many documented anomalies. Our analysis demonstrates that EBP not only survives stringent robustness tests but also meaningfully expands the investment opportunity set after accounting for trading costs—a criterion that eliminates nearly half of published anomalies. The finding that EBP maintains significant alphas when controlling for closely related signals from the factor zoo, including those based on debt financing and accruals studied in [Sloan \(1996\)](#), confirms that it captures distinct variation in expected returns.

These findings have broader implications for understanding how production frictions shape asset prices. The success of EBP in predicting returns supports theoretical models where firms’ investment decisions and financial policies jointly determine their risk exposures [Golanac and Miao \(2018\)](#). Our results suggest that investors can identify firms with different sensitivities to aggregate shocks by observing their equity preservation behavior, providing a practical implementation of production-based theories. The economic significance of the EBP premium, particularly among large-cap stocks where trading costs are minimal, indicates that production frictions remain first-order determinants of expected returns even in the most liquid segments of the equity market.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all non-financial firms with available data for the required variables over our sample period. We focus on annual observations to align with the reporting frequency of long-term debt issuance activities. Equity Buffer Preservation signal relies on two key accounting variables. Long-term debt issuance (DLTIS) rep-

resents the total proceeds from the issuance of long-term debt instruments during the fiscal year, capturing new borrowing activities with maturities exceeding one year. Capital surplus (CAPS) measures the amount shareholders have invested in the firm above the par value of the stock, including additional paid-in capital from stock issuances, stock-based compensation, and other equity transactions that exceed stated capital values. We construct the Equity Buffer Preservation signal by calculating the year-over-year change in long-term debt issuance scaled by lagged capital surplus: $(DLTIS(t) - DLTIS(t-1)) / CAPS(t-1)$. This measure captures the rate of change in a firm’s long-term borrowing activity relative to its existing equity cushion above par value. A higher value indicates accelerating debt issuance relative to the firm’s accumulated capital surplus, potentially signaling efforts to preserve equity buffers through increased reliance on debt financing. We compute this signal using fiscal year-end values and require non-missing values for DLTIS in consecutive years and non-zero CAPS in the lagged period. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles of its empirical distribution each year.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EBP signal. Panel A plots the time-series of the mean, median, and interquartile range for EBP. On average, the cross-sectional mean (median) EBP is -2.08 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input EBP data. The signal’s interquartile range spans -0.44 to 0.32. Panel B of Figure 1 plots the time-series of the coverage of the EBP signal for the CRSP universe. On average, the EBP signal is available for 5.68% of CRSP names, which on average make up 6.56% of total market capitalization.

4 Does EBP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EBP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EBP portfolio and sells the low EBP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EBP strategy earns an average return of 0.28% per month with a t-statistic of 3.77. The annualized Sharpe ratio of the strategy is 0.52. The alphas range from 0.22% to 0.31% per month and have t-statistics exceeding 2.96 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.23, with a t-statistic of 4.52 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 471 stocks and an average market capitalization of at least \$1,689 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 4.96. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 1-23bps/month. The lowest return, (1 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.13. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EBP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the EBP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EBP, as well as average returns and alphas for long/short trading EBP strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EBP strategy achieves an average return of 29 bps/month with a t-statistic of 3.28. Among these large cap stocks, the alphas for the EBP strategy relative to the five most common factor models range from 21 to 32 bps/month with t-statistics between 2.39 and 3.66.

5 How does EBP perform relative to the zoo?

Figure 2 puts the performance of EBP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EBP strategy falls in the distribution. The EBP strategy’s gross (net) Sharpe ratio of 0.52 (0.43) is greater than 92% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EBP strategy (red line).² Ignoring trading costs, a \$1 invested in the EBP strategy would have yielded \$3.57 which ranks the EBP strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EBP strategy would have yielded \$2.41 which ranks the EBP strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the EBP relative to those. Panel A shows that the EBP strategy gross alphas fall between the 57 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EBP strategy has a positive net generalized alpha for five out of the five factor models. In these cases EBP ranks between the 76 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EBP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of EBP with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EBP or at least to weaken the power EBP has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of EBP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBP}EBP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EBP. Stocks are finally grouped into five EBP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EBP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EBP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EBP signal in these Fama-MacBeth regressions exceed 0.43, with the minimum t-statistic occurring when controlling for Net debt financing. Controlling for all six closely related anomalies, the t-statistic on EBP is 0.40.

Similarly, Table 5 reports results from spanning tests that regress returns to the EBP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EBP strategy earns alphas that range from 18-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.48,

which is achieved when controlling for Net debt financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EBP trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.38.

7 Does EBP add relative to the whole zoo?

Finally, we can ask how much adding EBP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the EBP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes EBP grows to \$1108.07.

8 Conclusion

This study provides compelling evidence that Equity Buffer Preservation (EBP) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EBP-based trading strate-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EBP is available.

gies generate substantial risk-adjusted returns that persist even after accounting for transaction costs and controlling for well-established risk factors.

The key findings reveal that a value-weighted long/short strategy based on EBP delivers an annualized net Sharpe ratio of 0.43, which remains economically meaningful after accounting for implementation costs. More importantly, the signal’s alpha remains statistically significant at 18 basis points per month (t -statistic = 2.44) relative to the Fama-French five-factor model augmented with momentum, indicating that EBP captures unique information about expected returns not subsumed by conventional risk factors. The robustness of our results is further validated by the persistence of abnormal returns (17 bps/month, t -statistic = 2.38) even after controlling for six closely related factors from the factor zoo, including changes in financial liabilities, net debt financing, and accruals.

From a practical perspective, these findings suggest that EBP can serve as a valuable signal for portfolio construction and risk management. The strategy’s net Sharpe ratio of 0.43 indicates that it remains viable after realistic transaction costs, making it potentially implementable for institutional investors. The economic magnitude of the alpha, combined with its statistical significance, suggests that firms’ equity buffer preservation behaviors contain important information about future stock returns that market participants may not fully incorporate into prices.

However, several limitations warrant consideration. First, while our analysis controls for numerous established factors, the proliferation of anomalies in the academic literature raises concerns about data mining and the multiple testing problem. Second, the implementation of EBP strategies in practice may face capacity constraints and market impact costs beyond those captured in our transaction cost estimates. Third, the stability of the EBP signal across different market regimes and its performance during extreme market conditions require further investigation.

Future research should explore several promising avenues. First, investigating

the economic mechanisms underlying the EBP effect could provide deeper insights into why this signal predicts returns. Understanding whether EBP reflects risk compensation, behavioral biases, or institutional frictions would enhance our theoretical understanding of this anomaly. Second, examining the international evidence for EBP across different markets and regulatory environments would test the external validity of our findings. Third, analyzing the time-varying nature of the EBP premium and its relationship with macroeconomic conditions could reveal important dynamics that affect strategy performance. Finally, investigating the interaction between EBP and corporate financial policies, such as dividend payments and share repurchases, may uncover additional insights into how firms' capital structure decisions influence expected returns.

In conclusion, this paper establishes Equity Buffer Preservation as a robust and economically significant predictor of stock returns that survives stringent empirical tests and transaction cost adjustments. While questions remain about its underlying economic drivers and long-term persistence, the evidence suggests that EBP deserves serious consideration from both academic researchers and investment practitioners seeking to understand and exploit cross-sectional return predictability.

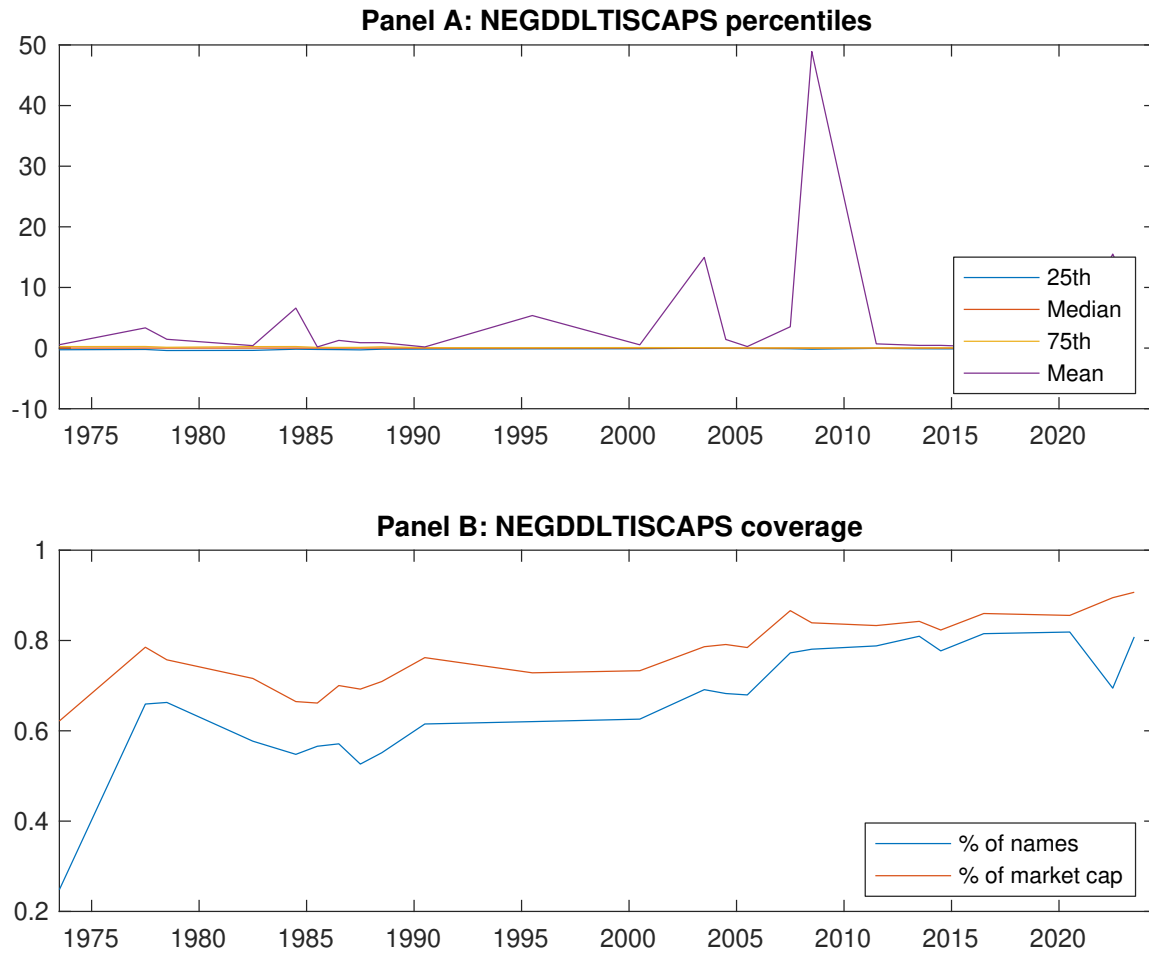


Figure 1: Times series of EBP percentiles and coverage.
This figure plots descriptive statistics for EBP. Panel A shows cross-sectional percentiles of EBP over the sample. Panel B plots the monthly coverage of EBP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EBP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on EBP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.59 [2.99]	0.65 [3.33]	0.61 [2.93]	0.68 [3.67]	0.86 [4.55]	0.28 [3.77]
α_{CAPM}	-0.07 [-1.30]	-0.02 [-0.32]	-0.08 [-1.17]	0.05 [1.05]	0.22 [4.27]	0.29 [3.98]
α_{FF3}	-0.11 [-2.09]	-0.04 [-0.75]	-0.01 [-0.08]	0.03 [0.74]	0.19 [3.75]	0.31 [4.17]
α_{FF4}	-0.09 [-1.61]	-0.01 [-0.26]	0.06 [0.99]	0.03 [0.59]	0.15 [2.93]	0.24 [3.26]
α_{FF5}	-0.18 [-3.22]	-0.02 [-0.40]	0.12 [1.88]	-0.00 [-0.09]	0.09 [1.81]	0.27 [3.58]
α_{FF6}	-0.15 [-2.78]	-0.00 [-0.08]	0.16 [2.62]	-0.00 [-0.11]	0.07 [1.33]	0.22 [2.96]
Panel B: Fama and French (2018) 6-factor model loadings for EBP-sorted portfolios						
β_{MKT}	1.04 [81.98]	1.02 [91.48]	0.98 [67.48]	0.98 [90.81]	1.03 [89.02]	-0.01 [-0.73]
β_{SMB}	-0.01 [-0.27]	-0.06 [-3.56]	-0.01 [-0.31]	0.00 [0.05]	0.06 [3.52]	0.07 [2.58]
β_{HML}	0.10 [4.02]	0.11 [5.12]	-0.16 [-5.69]	-0.01 [-0.36]	-0.02 [-1.09]	-0.12 [-3.72]
β_{RMW}	0.18 [7.12]	0.03 [1.32]	-0.19 [-6.56]	0.03 [1.53]	0.12 [5.17]	-0.06 [-1.80]
β_{CMA}	0.01 [0.15]	-0.11 [-3.25]	-0.19 [-4.38]	0.11 [3.47]	0.23 [6.85]	0.23 [4.52]
β_{UMD}	-0.04 [-2.89]	-0.02 [-2.17]	-0.07 [-4.99]	0.00 [0.13]	0.04 [3.30]	0.07 [4.38]
Panel C: Average number of firms (n) and market capitalization (me)						
n	486	560	1031	619	471	
me (\$10 ⁶)	1724	2516	1899	2445	1689	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EBP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [3.77]	0.29 [3.98]	0.31 [4.17]	0.24 [3.26]	0.27 [3.58]	0.22 [2.96]
Quintile	NYSE	EW	0.22 [4.96]	0.25 [5.50]	0.23 [5.22]	0.22 [4.75]	0.19 [4.38]	0.19 [4.17]
Quintile	Name	VW	0.24 [3.53]	0.27 [4.00]	0.27 [4.04]	0.21 [3.16]	0.21 [3.09]	0.17 [2.52]
Quintile	Cap	VW	0.23 [3.25]	0.26 [3.63]	0.27 [3.85]	0.22 [3.03]	0.24 [3.28]	0.20 [2.72]
Decile	NYSE	VW	0.24 [2.82]	0.28 [3.32]	0.27 [3.19]	0.23 [2.76]	0.20 [2.31]	0.18 [2.08]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [3.09]	0.23 [3.16]	0.24 [3.31]	0.21 [2.84]	0.21 [2.82]	0.18 [2.44]
Quintile	NYSE	EW	0.01 [0.13]	0.03 [0.56]	0.01 [0.25]	0.01 [0.17]		
Quintile	Name	VW	0.19 [2.81]	0.22 [3.26]	0.22 [3.28]	0.19 [2.83]	0.17 [2.49]	0.14 [2.14]
Quintile	Cap	VW	0.19 [2.59]	0.20 [2.84]	0.22 [3.02]	0.18 [2.57]	0.18 [2.53]	0.15 [2.18]
Decile	NYSE	VW	0.19 [2.22]	0.23 [2.73]	0.22 [2.64]	0.20 [2.42]	0.16 [1.88]	0.14 [1.71]

Table 3: Conditional sort on size and EBP

This table presents results for conditional double sorts on size and EBP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EBP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EBP and short stocks with low EBP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EBP Quintiles					EBP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.75 [2.89]	0.74 [2.53]	0.92 [3.35]	0.79 [2.64]	0.88 [3.23]	0.13 [1.32]	0.14 [1.50]	0.13 [1.33]	0.07 [0.73]	0.08 [0.86]	0.05 [0.47]
	(2)	0.78 [3.02]	0.85 [3.21]	0.88 [3.36]	0.85 [3.31]	0.86 [3.47]	0.08 [1.09]	0.11 [1.54]	0.09 [1.18]	0.07 [0.97]	0.06 [0.77]	0.05 [0.65]
	(3)	0.76 [3.14]	0.82 [3.46]	0.81 [3.26]	0.77 [3.28]	0.92 [3.93]	0.16 [2.07]	0.19 [2.54]	0.18 [2.41]	0.14 [1.85]	0.16 [2.05]	0.13 [1.67]
	(4)	0.69 [3.17]	0.90 [4.09]	0.81 [3.58]	0.77 [3.54]	0.82 [3.82]	0.12 [1.66]	0.14 [1.87]	0.13 [1.67]	0.10 [1.35]	0.09 [1.11]	0.07 [0.93]
	(5)	0.54 [2.81]	0.55 [2.82]	0.57 [2.86]	0.63 [3.27]	0.83 [4.47]	0.29 [3.28]	0.31 [3.45]	0.32 [3.66]	0.25 [2.83]	0.27 [2.95]	0.21 [2.39]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EBP Quintiles					EBP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	358	356	358	357	355	36	29	30	29	34	
	(2)	98	98	97	98	98	56	56	55	56	56	
	(3)	69	69	69	69	69	99	100	96	97	99	
	(4)	58	58	58	58	58	215	218	212	214	212	
(5)	53	52	53	52	53	1485	1882	1513	1898	1498		

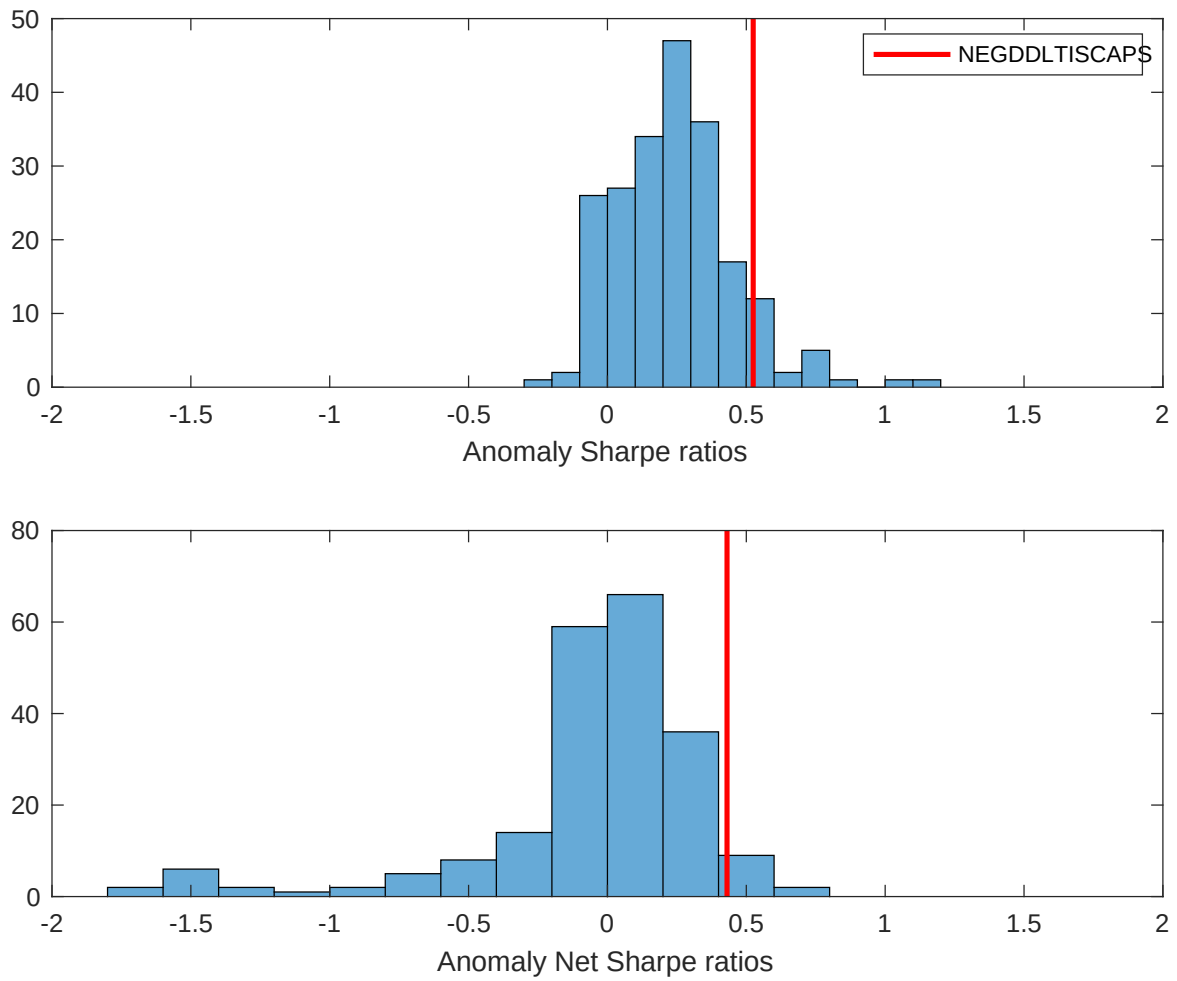


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EBP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

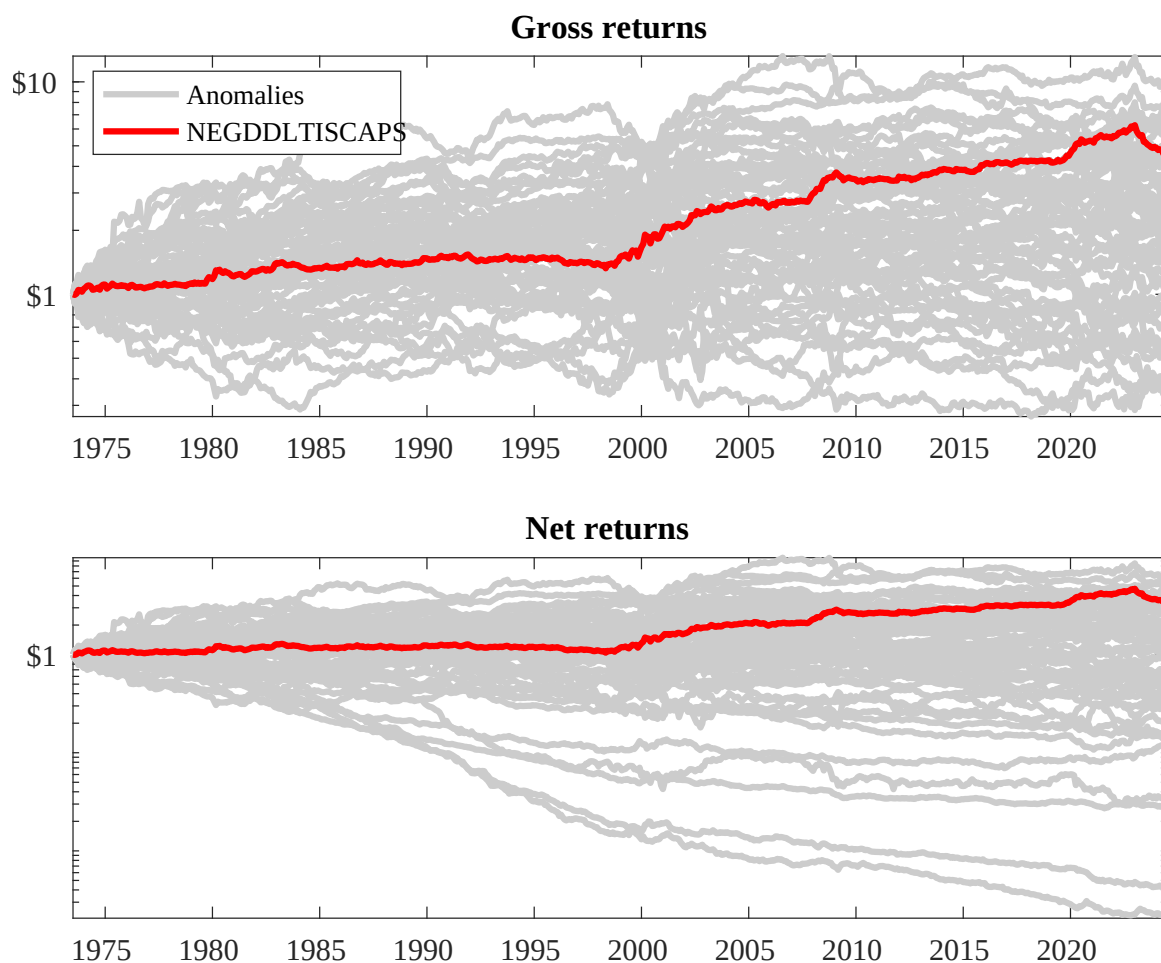
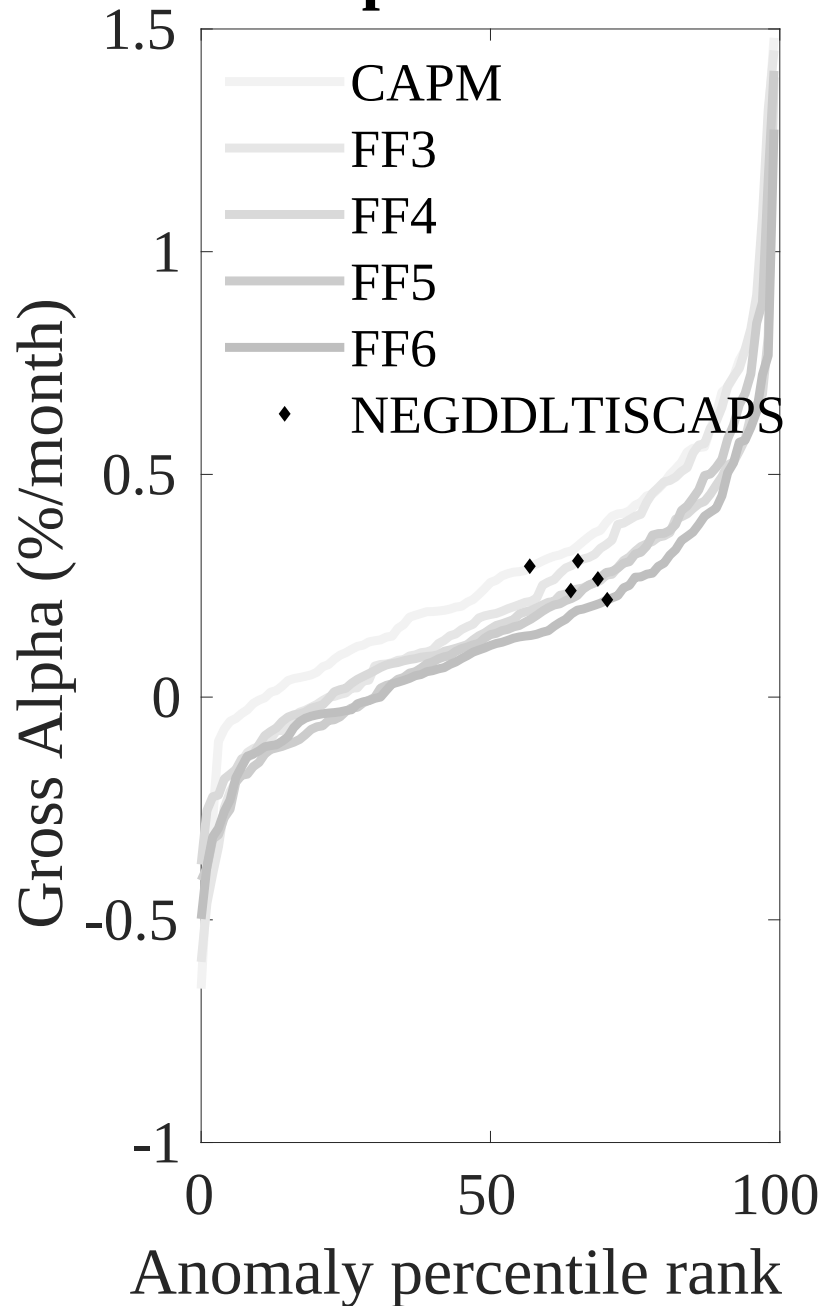


Figure 3: Dollar invested.

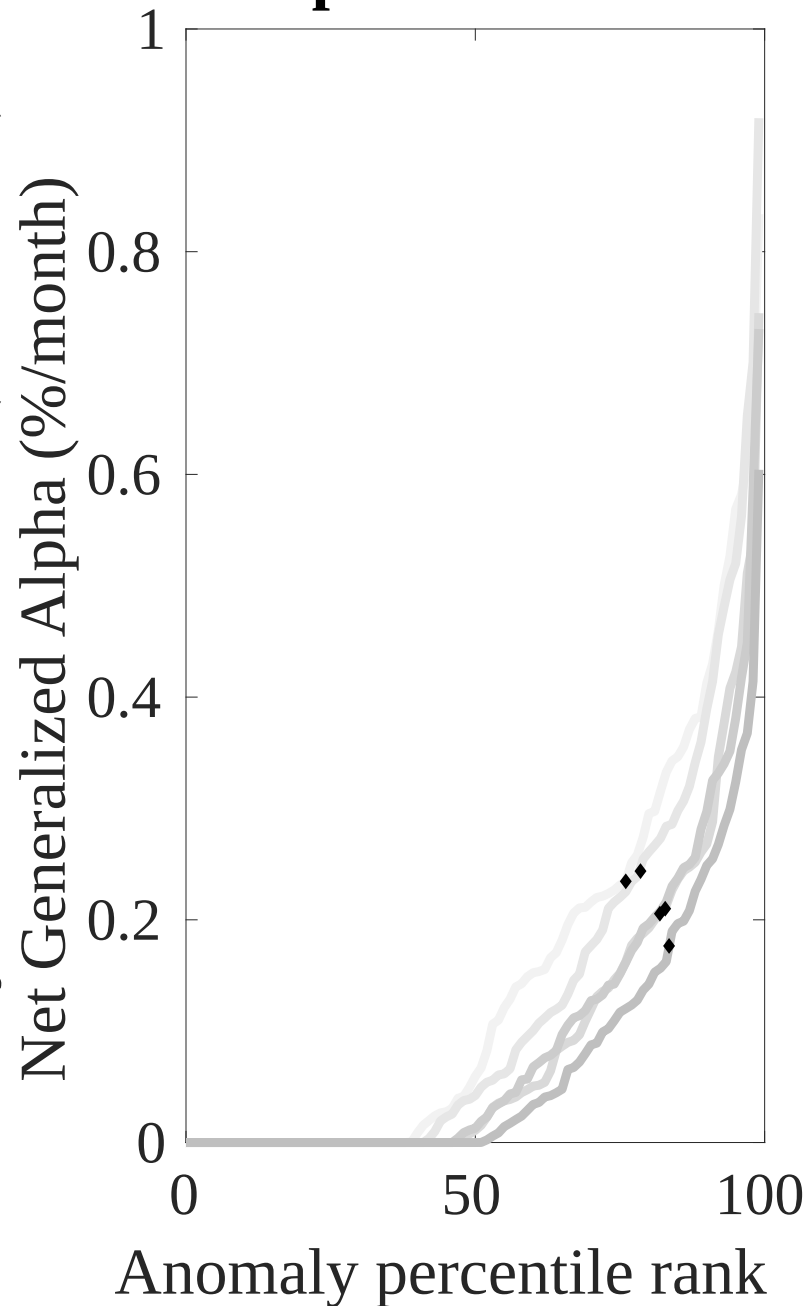
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EBP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



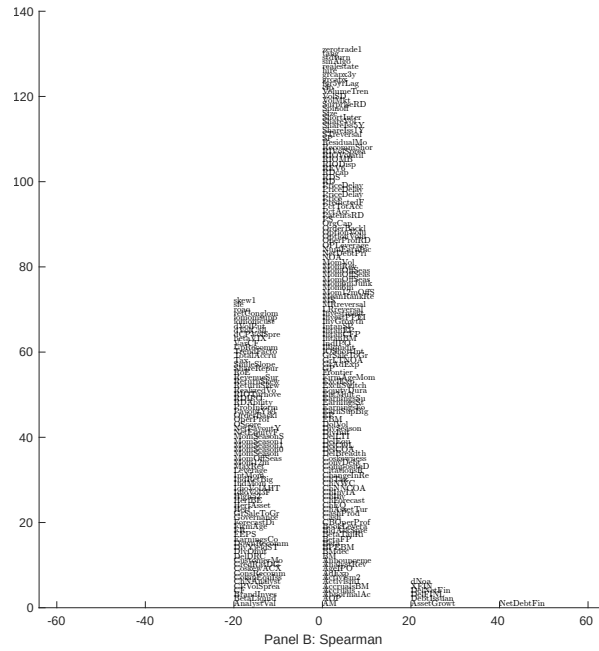
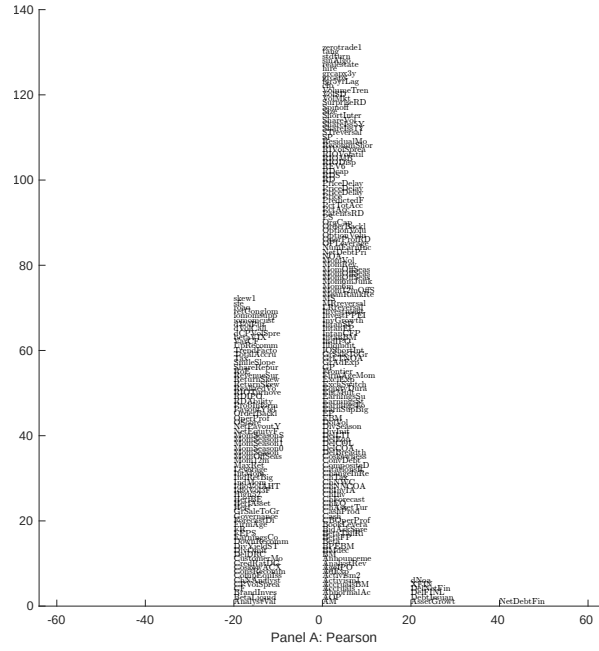


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EBP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

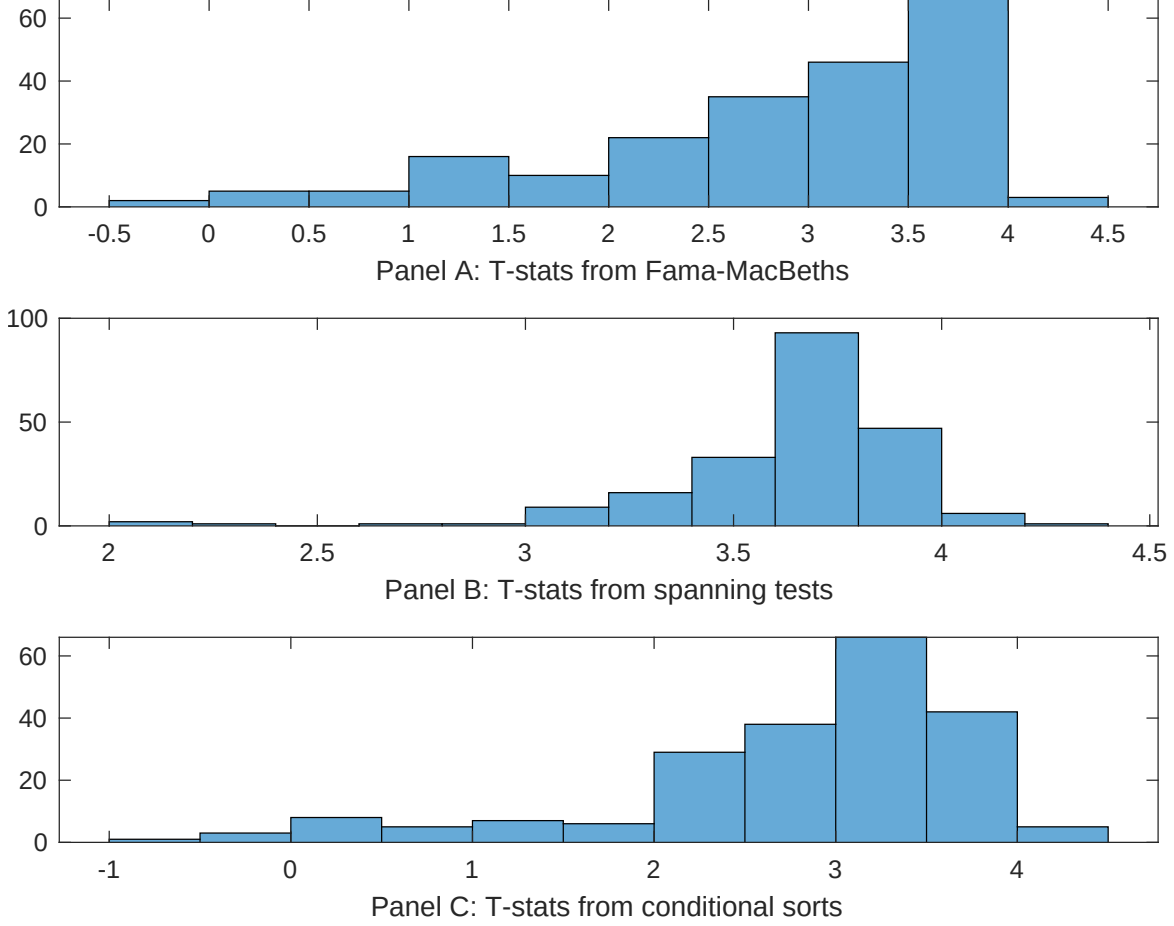


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EBP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBP}EBP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EBP. Stocks are finally grouped into five EBP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EBP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EBP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EBP}EBP_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Inventory Growth, Investment to revenue, Accruals, Book leverage (annual). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.28]	0.13 [5.25]	0.13 [5.22]	0.16 [6.27]	0.12 [4.92]	0.13 [5.19]	0.16 [6.47]
EBP	0.30 [0.53]	0.24 [0.43]	0.16 [2.48]	0.15 [3.11]	0.18 [3.42]	0.19 [3.69]	0.28 [0.40]
Anomaly 1	0.19 [9.45]						-0.18 [-0.04]
Anomaly 2		0.21 [8.96]					0.14 [2.73]
Anomaly 3			0.43 [7.25]				0.27 [4.08]
Anomaly 4				0.26 [5.99]			0.24 [4.60]
Anomaly 5					0.14 [4.38]		0.74 [1.80]
Anomaly 6						0.12 [1.27]	0.45 [0.38]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EBP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EBP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Inventory Growth, Investment to revenue, Accruals, Book leverage (annual). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.58]	0.19 [2.62]	0.20 [2.80]	0.19 [2.67]	0.19 [2.59]	0.18 [2.48]	0.17 [2.38]
Anomaly 1	12.32 [2.98]						-2.36 [-0.42]
Anomaly 2		14.59 [3.67]					16.87 [3.14]
Anomaly 3			4.82 [1.69]				4.59 [1.64]
Anomaly 4				6.12 [2.15]			4.97 [1.76]
Anomaly 5					3.85 [1.32]		-3.84 [-1.17]
Anomaly 6						8.99 [3.41]	11.61 [3.79]
mkt	-0.68 [-0.41]	-0.94 [-0.56]	-1.06 [-0.63]	-1.10 [-0.65]	-0.68 [-0.40]	0.15 [0.09]	-0.01 [-0.00]
smb	5.64 [2.19]	5.62 [2.20]	7.43 [2.90]	5.77 [2.22]	7.64 [2.93]	5.96 [2.34]	3.28 [1.19]
hml	-11.15 [-3.44]	-11.38 [-3.53]	-12.12 [-3.73]	-11.55 [-3.55]	-10.98 [-3.30]	-5.37 [-1.43]	-3.75 [-1.00]
rmw	-6.86 [-2.08]	-7.36 [-2.23]	-5.06 [-1.52]	-5.24 [-1.59]	-4.49 [-1.31]	-3.36 [-1.00]	-4.47 [-1.29]
cma	18.92 [3.71]	19.26 [3.85]	18.75 [3.38]	22.46 [4.55]	21.79 [4.34]	22.96 [4.69]	16.08 [2.86]
umd	6.53 [3.83]	6.65 [3.96]	7.10 [4.18]	6.81 [3.98]	7.31 [4.32]	7.41 [4.45]	5.64 [3.26]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	11	12	10	11	10	12	14

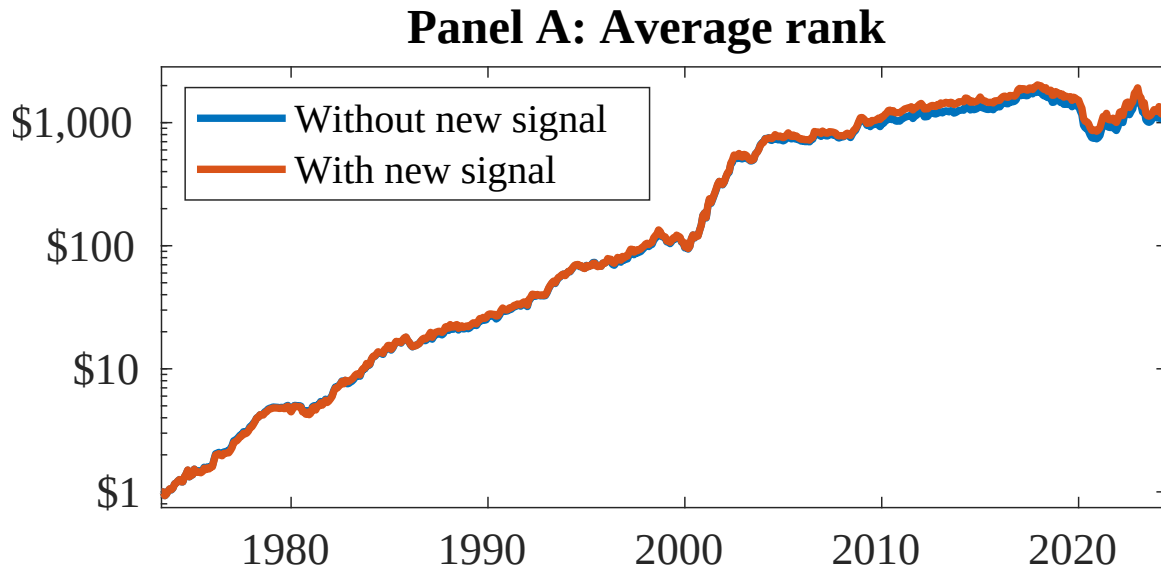


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as EBP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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